I decided to use a merge sort for the fourth week's assignment and then a BST for the Binary tree in week five.
I also implemented a pipeline that works that first it gets the JSON file and send it to ReadEmployeesTask that deserialises and then send it to InsertEmployeesIntoArrayTask, in there it copies the employee records into an employeeData[] Array. After this it gets split we have either the SortingUtils that is the merge sort for building up and sorting everyone. Then we have the EmployeeBinaryTreeTask that builds the tree and searches for the specific person we are looking for by searching for their ID. Finally it gets sent to ComputerTerminalUI that displays what JSON gets through and in what order.
I also used a Stopwatch through my TaskBenchmark.Mesure method to measure the time to execute the search. This let me see how long it would take if I needed to change anything for performance or if it gets stuck anywhere.

The reason why is for:

- Merge sort: Inside the SortingUtils I used a merge sort to sort through the 800 employees records on their entrance to the engine room. I implemented two binary searches through it, a LowerBound and an UpperBound to find each beginning and end of a specific time range and that can be changed inside the task in Unity.
- BST: For I already have a few queries and a different traversal technique from my earlier arrays. Also I picked the BST for the function of just searching for one signal individual with their ID. Also to demonstrate the tree's function.
  For we have 800 employees each with their own ID so to look for one person we have it look from employedata that gets its data from the JSON file. I also implemented Breadth-First Search BFS using Queue to traverse the tree level by level. I also added an Array.FindAll to filter out null employees and an Array.Sort to sort the employees ID.

Through this assignment I have not had that many issues other than small issues that were caused by my own stupidity. For example my scroll view that would cut off and stop after 500 entries, after a few hours of debugging and adding unnecessary code I found that it was just an issue in the inspector in unity where I had put what the scrollview would read incorrectly.
Other issues I had was that when I worked on the StationAI I had a contentContainer with a VerticalLayout container.
Another issue was the implementation of the Helper. The JSON that the helper uses for some reason refused to be a JSON file, I created it and then put it in and it would just turn into a text file, I solved it somehow by copying the employee JSON and replacing it with the helpers. Also inside it I put a choice container that gave me the issue that I struggled with changing the size of each button but later found the solution in the padding inside the Vertical Layout Group that builds up the container.
Otherwise I did not have too many issues but I did have issues with github itself that it doesn't want in the first weeks to work. It would not be sent to the correct repository. What it had done was it created a folder called github on my user and there was the repository while where i was it was in a folder inside documents. So I do have four versions of this project because of this issue.
There are a few things I would have wanted to do differently, first I underestimated the time for I have been ahead most weeks but then I have not focused on adding more features until the very end that causes some of them to be messy and not as well made as I would have wanted them to be. Sadly the investigation Panel that would function as the players main

hub for the game play did not get finished. The idea was to have multiple panels inside that would work like tabs where the player could jump around and look at clues and find connections between who the victim and who the killer is.
I would also have wanted to make that a handful of employees in the game you could call/text and get some option what to say to them. This would have functioned like an interrogation and later on if the player harassed too many employees or accused too many HR would step in and the player gets a game over.

Personally I really liked this assignment. I'll maybe sit with it more into the future.